MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 8)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 20**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

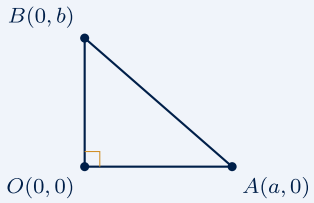
MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The vertices of a triangle are $O(0,0)$, $A(a,0)$ and $B(0,b)$. The coordinates of its circumcentre are: [1]

(A) (a,b) **(B)** $\left(\frac{a}{2},\frac{b}{2}\right)$ **(C)** $\left(\frac{a}{3},\frac{b}{3}\right)$ **(D)** $\left(\frac{a}{4},\frac{b}{4}\right)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $\left(\frac{a}{2},\frac{b}{2}\right)$



Given: Vertices $O(0,0)$, $A(a,0)$ on the x-axis, and $B(0,b)$ on the y-axis. Thus $\triangle OAB$ is a right-angled triangle with $\angle O = 90^\circ$ and hypotenuse AB.

We know that the circumcentre of a right-angled triangle is always the **midpoint of its hypotenuse**:

Formula Used: Circumcentre $C = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}\right)$

$$C = \left(\frac{a+0}{2}, \frac{0+b}{2}\right) = \boxed{\left(\frac{\mathbf{a}}{\mathbf{2}}, \frac{\mathbf{b}}{\mathbf{2}}\right)}$$

(Verification: $CO = CA = CB = \frac{\sqrt{a^2+b^2}}{2}$.)

Q2. A circle with centre $P(4,5)$ passes through the point $A(0,9)$. The length of the diagonal of the largest square inside this circle is: [1]

(A) $4\sqrt{2}$ units **(B)** $8\sqrt{2}$ units
(C) $\sqrt{53}$ units **(D)** $2\sqrt{53}$ units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $8\sqrt{2}$ **units**

Given: Centre $P(4,5)$ passes through $A(0,9)$.

Formula

Used: $r = PA = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$, Diagonal of largest inscribed square = Diameter = $2r$

$$r = \sqrt{(0-4)^2+(9-5)^2} = \sqrt{(-4)^2+4^2} = \sqrt{16+16} = \sqrt{32} = 4\sqrt{2} \text{ units}$$

The diagonal of the largest square inscribed in the circle equals its diameter:

$$\text{Diagonal} = 2r = 2 \times 4\sqrt{2} = \boxed{\mathbf{8\sqrt{2}} \text{ units}}$$

Q3. If the points $A(1,2)$, $O(0,0)$ and $C(a,b)$ are collinear, then: [1]

(A) $a = b$ **(B)** $a = 2b$ **(C)** $2a = b$ **(D)** $a = -b$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $2a = b$

Given: $A(1,2)$, $O(0,0)$, and $C(a,b)$ are collinear.

Formula Used: By Section Formula, if O(0,0) divides AC in ratio k:1, then ka+1=0 and kb+2=0.

If $a = 0$, $O = (0,0)$ forces $C = (0,0) \implies b = 0$ $(2a = b)$. For $a \neq 0$:

$$k = -\frac{1}{a} \implies \left(-\frac{1}{a}\right) b + 2 = 0 \implies \frac{b}{a} = 2 \implies \boxed{\mathbf{2a = b}}$$

Q4. $AOBC$ is a rectangle whose three vertices are $A(0,2)$, $O(0,0)$ and $B(4,0)$. The square of the length of its diagonal is equal to: [1]

(A) 36 **(B)** 20 **(C)** 16 **(D)** 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 20

Given: Rectangle $AOBC$ with $A(0,2)$, $O(0,0)$, $B(4,0)$. Here AB is one of its two equal diagonals.

Formula Used: $AB^2 = (x_2-x_1)^2+(y_2-y_1)^2$

$$AB^2 = (4-0)^2+(0-2)^2 = 4^2+(-2)^2 = 16+4 = \boxed{\mathbf{20}}$$

Q5. The distance of the point $P(-3,-4)$ from the origin is: [1]

(A) 3 **(B)** 4 **(C)** 5 **(D)** 25

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 5

Given: Point $P(-3,-4)$ and origin $O(0,0)$.

Formula Used: $OP = \sqrt{x^2+y^2}$

$$OP = \sqrt{(-3)^2+(-4)^2} = \sqrt{9+16} = \sqrt{25} = \boxed{\mathbf{5}\text{ units}}$$

SECTION B: Short Answer Type Questions

MATHS MANTHAN | (6 Marks)

Q6. In what ratio does the point $(2,p)$ divide the line segment joining the points $(-4,2)$ and $(5,-1)$? Also, find the value of 'p'. [2]

— OR —

The points $A(3,6)$, $B(k,2)$ and $C(6,2)$ are the vertices of a right triangle ABC right-angled at B. Find the value of k. [2]

▶ **SOLUTION (OPTION I):** [Total: 2 Marks]



Step 1: Finding the Internal Ratio [1 Mark]

Given: Endpoints $A(-4,2)$ and $B(5,-1)$, and dividing point $P(2,p)$. Let P divide segment AB internally in the ratio $r:1$.

We know the internal section formula:

Formula Used: $P(x,y) = \left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2}, \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

Equating the abscissa to 2:

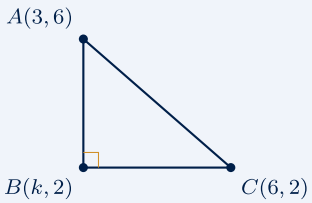
$$\frac{r(5)+1(-4)}{r+1} = 2 \implies 5r-4 = 2r+2 \implies 3r = 6 \implies r = 2 \implies \text{Ratio} = \mathbf{2:1}$$

Step 2: Finding the Value of p [1 Mark]

$$p = \frac{2(-1)+1(2)}{2+1} = \frac{-2+2}{3} = \frac{0}{3} = \mathbf{0}$$

$$\boxed{\text{Ratio } = \mathbf{2:1} \quad \text{and} \quad \mathbf{p = 0}}$$

▶ **SOLUTION (OPTION II):** [OR: 2 Marks]



Step 1: Applying Pythagoras' Theorem at Right Angle B [1 Mark]

Given: $\triangle ABC$ is right-angled at B, with vertices $A(3,6)$, $B(k,2)$, and $C(6,2)$. By Pythagoras' theorem, $AB^2 + BC^2 = AC^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Step 2: Solving for k and Checking Non-Degeneracy [1 Mark]

$$(k-3)^2 + (2-6)^2 + (6-k)^2 + (2-2)^2 = (6-3)^2 + (2-6)^2$$

$$(k-3)^2 + 16 + (6-k)^2 + 0 = 9 + 16$$

$$(k^2 - 6k + 9) + (36 - 12k + k^2) = 9 \implies 2k^2 - 18k + 36 = 0 \implies k^2 - 9k + 18 = 0$$

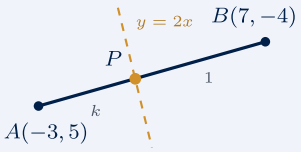
$$(k-3)(k-6) = 0 \implies k = 3 \quad \text{or} \quad k = 6$$

Note that for $k = 6$, point $B(6,2)$ coincides with $C(6,2)$ ($BC = 0$), which degenerates the triangle. Hence the unique valid right triangle occurs at:

$$\boxed{\mathbf{k = 3}}$$

Q7. A point P divides the line segment joining the points $A(-3,5)$ and $B(7,-4)$ in a certain ratio. If the point P lies on the line $y = 2x$, then find the ratio $AP : PB$ and coordinates of point P. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Assuming Ratio $k:1$ and Substituting into $y = 2x$ [1 Mark]

Given: Endpoints $A(-3,5)$ and $B(7,-4)$, and dividing point $P(x,y)$ lies on the line $y = 2x$. Let $AP : PB = k : 1$.

We know the internal section formula:

Formula Used: $P(x,y) = \left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$

Thus $P = \left(\dfrac{7k-3}{k+1}, \dfrac{-4k+5}{k+1}\right)$. Since P satisfies $y = 2x$:

$$\frac{-4k+5}{k+1} = 2\left(\frac{7k-3}{k+1}\right) \implies -4k + 5 = 14k - 6 \implies 11 = 18k \implies k = \frac{11}{18}$$

Step 2: Coordinates of P [1 Mark]

Using $m_1 : m_2 = 11 : 18$:

$$x = \frac{11(7) + 18(-3)}{11 + 18} = \frac{77 - 54}{29} = \frac{23}{29}, \qquad y = \frac{11(-4) + 18(5)}{11 + 18} = \frac{-44 + 90}{29} = \frac{46}{29}$$

$$\boxed{AP : PB = \mathbf{11 : 18} \quad \text{and} \quad P = \left(\mathbf{\frac{23}{29}}, \mathbf{\frac{46}{29}}\right)}$$

Q8. Find the coordinates of the midpoint of the line segment joining the points $(2, 3)$ and $(4, 7)$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Declaration [0.5 Mark]

Given: Endpoints $A(2, 3)$ and $B(4, 7)$.

Formula Used: Midpoint $M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

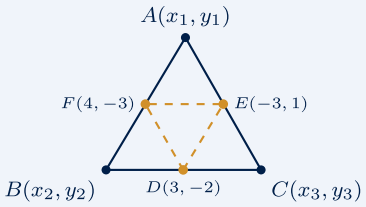
Step 2: Substitution and Evaluation [1.5 Marks]

$$M(x, y) = \left(\frac{2+4}{2}, \frac{3+7}{2}\right) = \left(\frac{6}{2}, \frac{10}{2}\right) = \boxed{(3, 5)}$$

SECTION C: Long Answer & Competency Based Questions MATHS MANTHAN | (9 Marks)

Q9. Let $D(3, -2)$, $E(-3, 1)$, $F(4, -3)$ be mid-points of sides BC, CA, AB respectively of $\triangle ABC$. Find co-ordinates of its vertices A, B and C. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Midpoint Equations for Sides [1.5 Marks]

Let vertices be $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$. Here $D(3, -2)$, $E(-3, 1)$, and $F(4, -3)$ are midpoints of sides BC, CA, and AB respectively.

We know the midpoint formula:

Formula Used: Midpoint $= \left(\frac{x_i + x_j}{2}, \frac{y_i + y_j}{2}\right)$

From the midpoint equations:

$$x_2 + x_3 = 2(3) = 6, \qquad x_3 + x_1 = 2(-3) = -6, \qquad x_1 + x_2 = 2(4) = 8$$

$$y_2 + y_3 = 2(-2) = -4, \qquad y_3 + y_1 = 2(1) = 2, \qquad y_1 + y_2 = 2(-3) = -6$$

Step 2: Solving for Abscissae x_1, x_2, x_3 [1.75 Marks]

Adding all three x-equations:

$$2(x_1 + x_2 + x_3) = 6 + (-6) + 8 = 8 \implies x_1 + x_2 + x_3 = 4$$

Subtracting each pair gives:

$$x_1 = 4 - (x_2 + x_3) = 4 - 6 = \mathbf{-2}, \quad x_2 = 4 - (-6) = \mathbf{10}, \quad x_3 = 4 - 8 = \mathbf{-4}$$

Step 3: Solving for Ordinates y_1, y_2, y_3 [1.75 Marks]

Adding all three y-equations:

$$2(y_1 + y_2 + y_3) = -4 + 2 + (-6) = -8 \implies y_1 + y_2 + y_3 = -4$$

Subtracting each pair gives:

$$y_1 = -4 - (-4) = \mathbf{0}, \quad y_2 = -4 - 2 = \mathbf{-6}, \quad y_3 = -4 - (-6) = \mathbf{2}$$

$$\boxed{\text{Vertices are } \mathbf{A(-2, 0)}, \mathbf{B(10, -6)}, \text{ and } \mathbf{C(-4, 2)}}$$

Q10. Competency Case Study: Trees act as natural filters. By planting trees in and around school premises, we create cleaner and healthier air for students and local residents, reducing respiratory problems. A school in Noida has proposed and organised a community drive on tree plantation under the title "Save Earth, Plant Trees". Students of that school have planted saplings in the field such that it formed a quadrilateral $ABCD$ as shown in the figure. *(From the Cartesian Grid diagram: $A(-5, 9)$, $B(2, 3)$, $C(1, -5)$, $D(-6, 1)$, and intersection of diagonals $P(-2, 2)$)* **[4]**

(i) Find the distance between the two saplings at A and D. **[1 Mark]**

(ii) **(a)** One student plants one sapling at the mid-point of AD. Then he moves along a straight line parallel to DB and sows another sapling on AB. What are the coordinates of the positions of these two new saplings? **[2 Marks]**

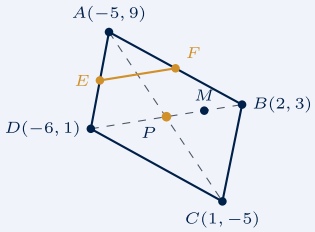
— OR —

(iii) **(b)** A new sapling is kept at a point M on DB such that $DM : MB = 3 : 1$. Find the coordinates of M. **[2 Marks]**

(iv) The line segments AC and BD bisect each other at $P(-2, 2)$. Find the coordinates of C. **[1 Mark]**

MATHS MANTHAN TUITION CLASSES OFFICIAL SOLUTIONS

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Given: Saplings at $A(-5,9)$, $B(2,3)$, $C(1,-5)$, and $D(-6,1)$, with diagonals AC and BD bisecting at $P(-2,2)$. We apply the distance, midpoint, and internal section formulas.

We know the distance and internal section formulas:

Formula Used:
$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$P(x,y) = \left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}\right)$$

(i) Distance Between $A(-5,9)$ and $D(-6,1)$: [1 Mark]

$$AD = \sqrt{(-6-(-5))^2+(1-9)^2} = \sqrt{(-1)^2+(-8)^2} = \sqrt{1+64} = \boxed{\sqrt{\mathbf{65}}\text{ units}}$$

(ii) (a) Positions of the Two New Saplings on AD and AB: [2 Marks]

Let E be the midpoint of AD ($A(-5,9)$, $D(-6,1)$):

$$E = \left(\frac{-5+(-6)}{2}, \frac{9+1}{2}\right) = \left(-\frac{\mathbf{11}}{\mathbf{2}}, \mathbf{5}\right) = (-5.5,\ 5)$$

In $\triangle ADB$, a line through the midpoint E of AD parallel to side DB bisects the other side AB at F (by Converse of Midpoint Theorem). So F is the midpoint of AB ($A(-5,9)$, $B(2,3)$):

$$F = \left(\frac{-5+2}{2}, \frac{9+3}{2}\right) = \left(-\frac{\mathbf{3}}{\mathbf{2}}, \mathbf{6}\right) = (-1.5,\ 6)$$

— OR —

(iii) (b) Coordinates of M on DB ($DM : MB = 3:1$): [OR: 2 Marks]

Using the internal section formula for $D(-6,1)$ and $B(2,3)$ in ratio $3:1$:

$$M = \left(\frac{3(2)+1(-6)}{3+1}, \frac{3(3)+1(1)}{3+1}\right) = \left(\frac{6-6}{4}, \frac{9+1}{4}\right) = \left(0, \frac{10}{4}\right) = \left(\mathbf{0}, \frac{\mathbf{5}}{\mathbf{2}}\right) = (0,\ 2.5)$$

(iv) Finding Coordinates of C Using Diagonal Bisection at $P(-2,2)$: [1 Mark]

Since $P(-2,2)$ is the midpoint of diagonal AC with $A(-5,9)$ and $C(x_C, y_C)$:

$$\frac{-5+x_C}{2} = -2 \implies -5+x_C = -4 \implies x_C = 1$$
$$\frac{9+y_C}{2} = 2 \implies 9+y_C = 4 \implies y_C = -5$$

$\boxed{(i)\ \sqrt{65}\text{ units};\ \ (a)\ (-5.5, 5),\ (-1.5, 6)\ [\text{OR } (b)\ (0, 2.5)];\ \ (iv)\ C(1,-5)}$

MATHS MANTHAN • OFFICIAL SOLUTION

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★